

Class activity: inference for the regression slope

Group members:

Instructions: Work with a neighbor on the following activity. I will collect the handout at the end of class, and it will be part of your class participation grade. You will be graded only on effort – it is ok if you don't finish all the questions, or get them all correct.

Sparrow data

In this activity, you will practice hypothesis testing with a linear regression model. We will work with data on 116 sparrows, which were observed on Kent Island, New Brunswick. For each sparrow, we have information including:

- *Weight*: the weight of the sparrow (in grams)
- *WingLength*: the sparrow's wing length (in mm)

We fit the model

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

Here is output from the fitted model:

Coefficients:

	Estimate	Std. Error
(Intercept)	8.75465	1.39601
Weight	1.31341	0.09756

Questions

Researchers want to know whether there is a relationship between sparrow weight and wing length.

1. Write down null and alternative hypotheses corresponding to the research question.

